

JESS Thermodynamic Database v8.9

Lawrencium

24-Jan-23

Reaction No. 71085							
$\text{Lr}+3 + 3\text{e-1} = \text{Lr(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.96(3SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:-1.96(3SF)	3	CRV	7761
3	25	0	Inf. Dilution	E:-2.1(2SF)	0	CRV	22551
4	25	0	Inf. Dilution	dH:599.3(4SF) kJ	3	CRV	7761

Reaction No. 71652							
$\text{Lr}+3_{\text{OH-1}(3)_{\text{(am.,s)}}} + 3\text{e-1} = \text{Lr(s)} + 3\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.46(3SF)	3	CRV	7761

Reaction No. 71653							
$\text{Lr}+3_{\text{OH-1}(3)_{\text{(s)}}} + 3\text{e-1} = \text{Lr(s)} + 3\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.49(3SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:636.2(4SF) kJ	3	CRV	7761

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)

1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review

References

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A
Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 7761(89Bra)

J. Phys. Chem. Ref. Data, 1989, 18, 1 - 21; Bratsch SG; Standard Electrode Potentials
and Temperature Coefficients in Water at 298.15 K

Reference No.: 22551(06Inz)

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and Other Characteristic Potentials of Selected Electrode Reactions